

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

Smart School Management System — RBAC Scenario

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1. Epics

The following five Epics were created in Jira, breaking down the Smart School Management System's functional requirements (from Lab 1) into major themes:

Epic 1: Student Academic Profile Management

Enable students to securely view their own profile, academic history, class/section assignment, attendance records, and fee status.

Epic 2: Teacher Attendance & Classroom Management

Enable teachers to mark and manage student attendance, and access authorized student and personal information relevant to their assigned classes.

Epic 3: Admin User & Role Management (RBAC)

Enable admins to perform full CRUD operations on users, manage roles and permissions, and oversee fee and salary records across the system.

Epic 4: Principal Read-Only Oversight Dashboard

Enable principals/headmasters to view read-only, aggregated data across students, teachers, attendance, and academic records.

Epic 5: Fees & Salary Processing

Enable secure recording, tracking, and processing of student fee payments and staff salary disbursements, with historical records preserved per academic year.

Backlog View — Epics and User Stories (before estimation)

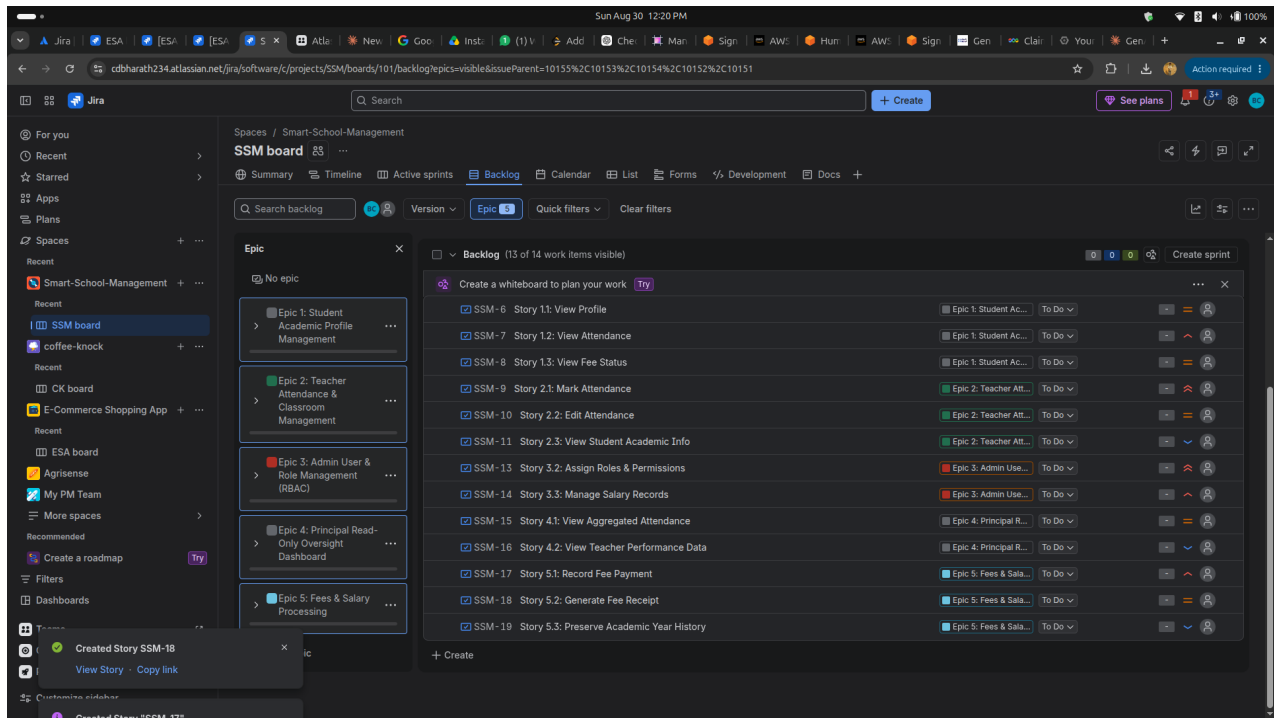


Figure 1: Full backlog showing all 5 Epics and their linked User Stories, each in the To Do status, with priority flags assigned (arrows indicate High/Highest or Low priority).

2. Story Point Estimation

Story points were assigned to each User Story using the Fibonacci scale (0, 1, 2, 3, 5, 8, 13...), reflecting relative complexity, effort, and uncertainty.

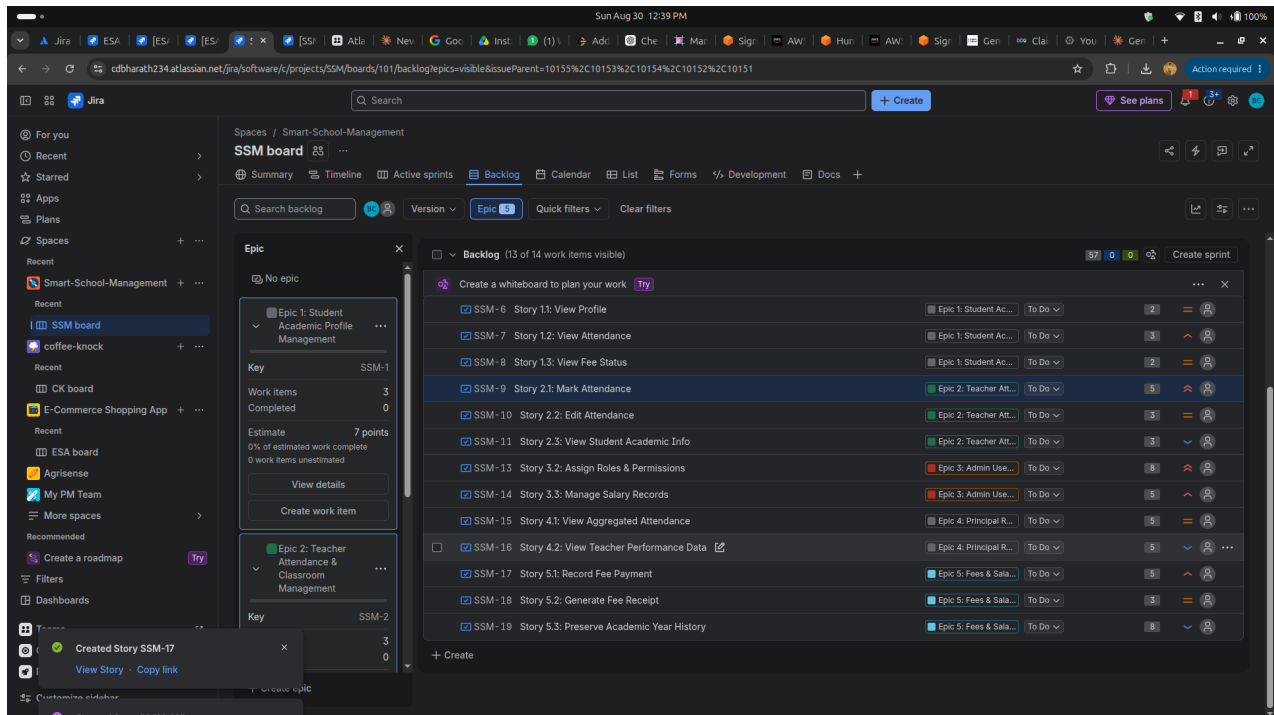


Figure 2: Backlog with Story Points assigned to each item (visible per-row) and the running Epic estimate total (e.g. Epic 1 shows 7 points across 3 stories). The visible backlog items sum to 57 total story points.

Note: 13 of 14 tracked work items are visible in this filtered view. One item (likely Story 3.1: Create User Accounts, under Epic 3) does not appear in the backlog screenshots — worth double-checking it was actually created and estimated before final submission.

3. Sprint Execution

Sprint 1 — Active Sprint Board

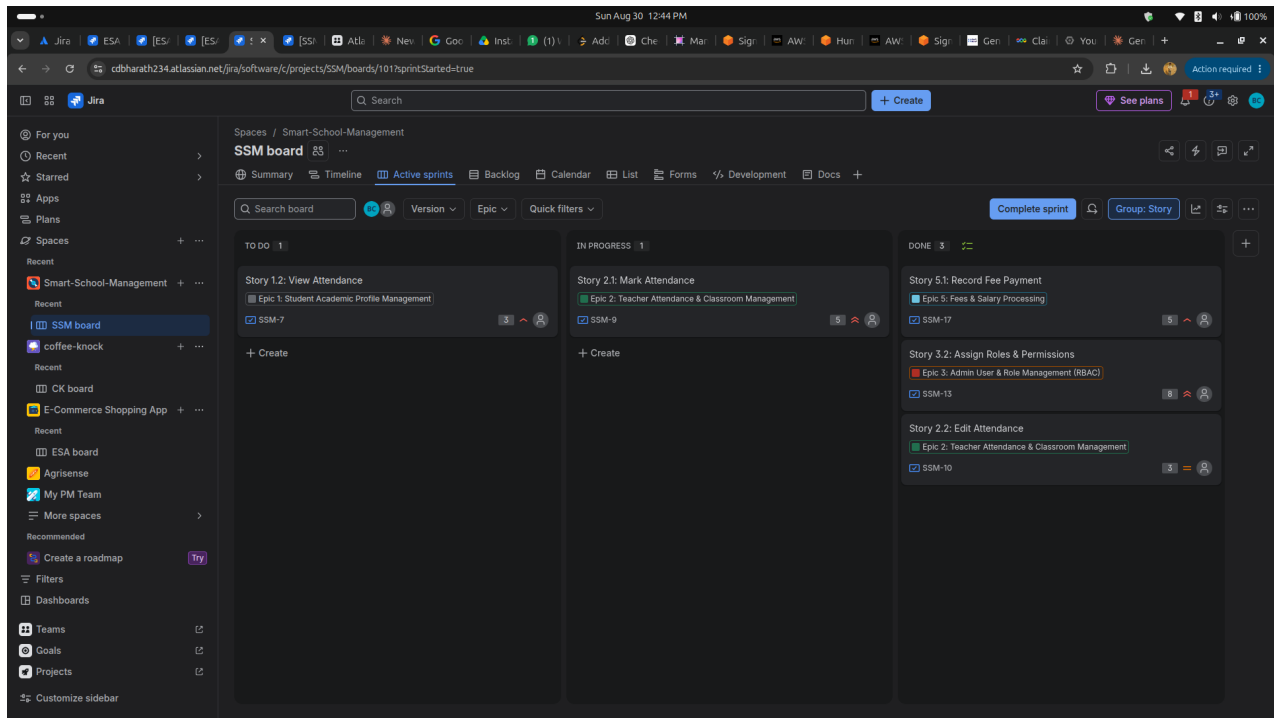


Figure 3: Sprint 1 in progress. Of 5 stories pulled into the sprint (24 story points total), 3 stories reached Done — Story 5.1: Record Fee Payment (5 pts), Story 3.2: Assign Roles & Permissions (8 pts), and Story 2.2: Edit Attendance (3 pts), totaling 16 points completed. Story 2.1: Mark Attendance (5 pts) remained In Progress and Story 1.2: View Attendance (3 pts) remained in To Do.

Sprint 1 Completed → Sprint 2 Auto-Created with Rollover Items

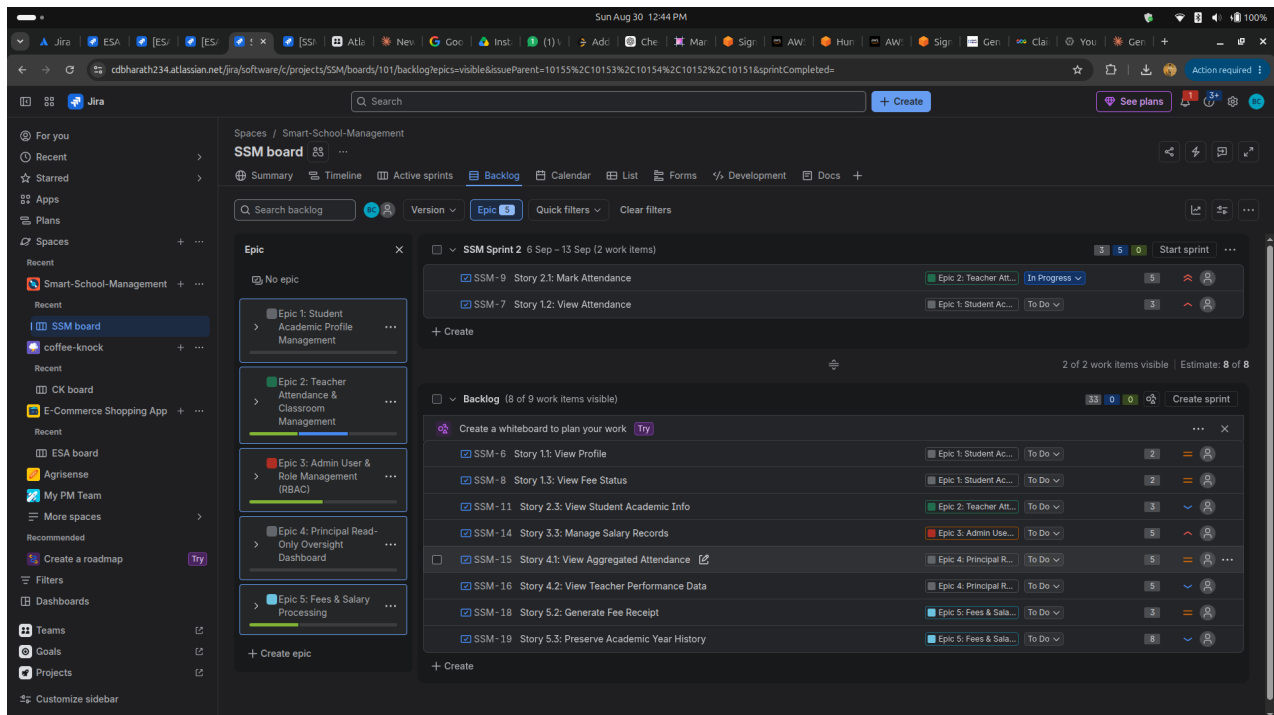


Figure 4: After completing Sprint 1, its two unfinished stories (Story 2.1 and Story 1.2, totaling 8 points) automatically rolled over into a new Sprint 2. The remaining backlog (8 more unstarted stories) is visible below, along with per-epic progress bars showing partial completion for Epics 2, 3, 4, and 5.

Sprint 2 — Completed

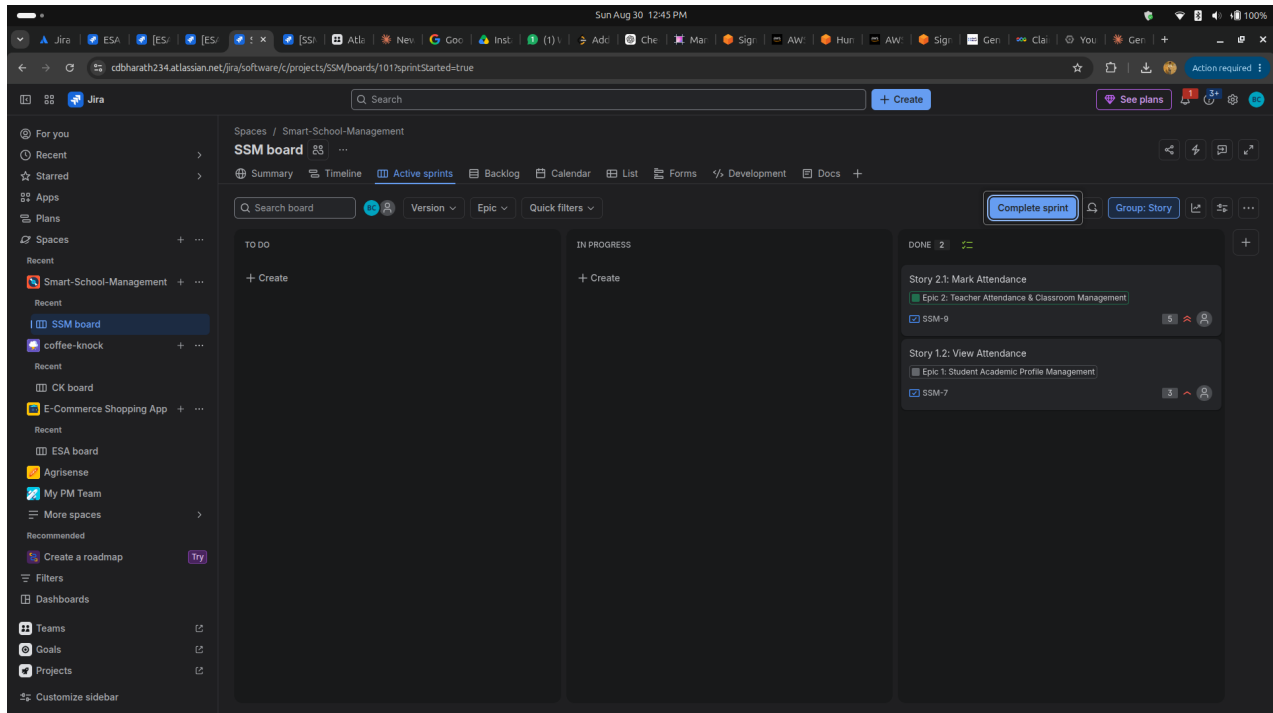


Figure 5: Sprint 2 board just before completion — both rolled-over stories (Story 2.1: Mark Attendance, 5 pts; Story 1.2: View Attendance, 3 pts) reached Done, giving Sprint 2 a 100% completion rate (8 of 8 points).

4. Reports Overview

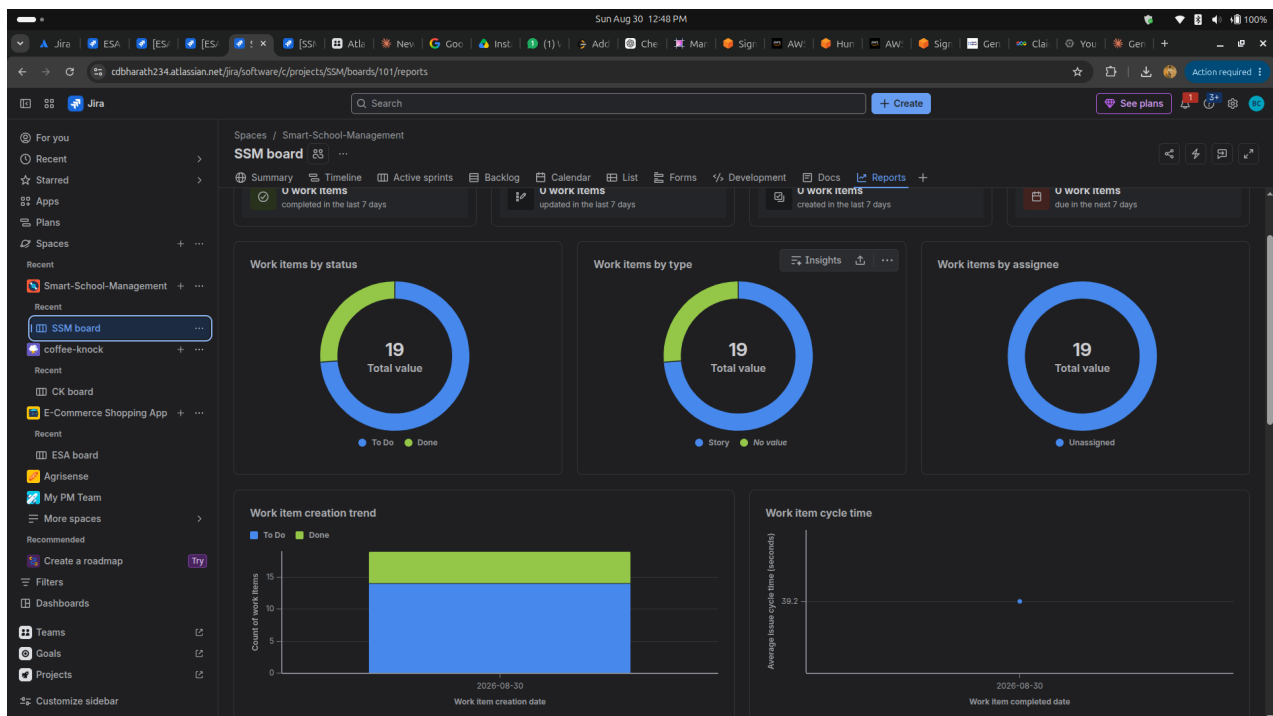


Figure 6: Jira's Reports overview for the project — 19 total work items, split between To Do and Done status, all typed as Story, all Unassigned, with a work item creation trend and an average cycle time of ~39.2 seconds (a result of simulating the sprint timeline in minutes rather than across real days).

Note: this screenshot shows the general work-item Reports overview rather than the classic sprint Burndown Chart (Remaining Values vs. Guideline line graph) specifically requested in the lab handout. It's worth navigating to Reports → Burndown Chart directly and capturing that view as well, since the handout lists the Burndown Chart as one of the two required PDF deliverables.

5. Reflection Questions

1. Did your estimations reflect the actual effort?

Largely, yes. The highest-effort story in the backlog, Story 3.2: Assign Roles & Permissions (8 points, correctly flagged as core RBAC logic), was completed within Sprint 1 despite its size, suggesting the point value didn't block progress when prioritized early. However, Story 2.1: Mark Attendance (5 points) took longer than expected and remained In Progress at the end of Sprint 1, rolling into Sprint 2 — indicating its complexity (daily workflow logic) may have been slightly underestimated relative to similarly-pointed stories like Story 3.3 (5 points) which wasn't even started yet. Overall the Fibonacci spread (2/3/5/8) matched the relative shape of the work, but one 5-point story clearly took more real effort than another 5-point story.

2. Was your backlog well-prioritized?

Partially. Story 3.2 (High priority, 8 points) and Story 5.1 (High priority, 5 points) were completed in Sprint 1, which aligns with prioritizing high-impact RBAC and fee-payment logic early. However, Story 2.1: Mark Attendance was flagged Highest priority but still wasn't finished by the end of Sprint 1, and Story 1.2: View Attendance (High priority) wasn't even started — both had to roll over. This suggests the sprint may have taken on slightly more Highest/High priority work than the team's simulated capacity could actually finish in the timebox, or that lower-priority quick wins (like Story 2.2, Medium priority) were tackled ahead of some higher-priority items.

3. How did your simulated sprint align with your plan?

Sprint 1 planned 5 stories worth 24 story points but only completed 3 stories (16 points) by the end of the timebox — a 67% completion rate by points. The 2 incomplete stories (8 points combined) automatically carried over into Sprint 2, which then completed both remaining stories for a 100% completion rate. This two-sprint pattern is a realistic reflection of real sprint planning: the first sprint was slightly over-committed, and the rollover mechanism let the team recover and finish the leftover work in the very next sprint rather than losing track of it.

4. What insights did the burndown chart give about your team's capacity?

The Reports overview shows 19 total work items split between To Do and Done status, with all items unassigned and typed as Story — useful context but not a direct capacity signal on its own. The rollover from Sprint 1 to Sprint 2 (2 of 5 stories, or roughly a third of committed points, carrying over) is the clearer capacity signal here: it suggests the team's real throughput per sprint is closer to 16–20 points than the 24 points initially committed in Sprint 1. For future sprint planning, committing to a slightly smaller batch of stories per sprint (closer to the actual completed velocity) would likely produce a more accurate forecast.

End of report. Deliverables included: Epics (Section 1), Story Points (Section 2), Sprint boards (Section 3), Reports/Burndown evidence (Section 4), and Reflection Questions (Section 5).